

Term 3 — Lesson 1 Practice Activity 1 (C1–C3) with Solutions

Contents

General problem	1
1 Daily app check-ins for students and recent graduates	3
2 Evening study time during a demanding term	4
3 Defect rate in portable chargers	6
4 Homework completion time in an evening certification program	7

Evaluated criteria

- C1: Describe the concept of a statistical hypothesis in context.
- C2: Identify the null and alternative hypotheses using correct parameter notation.
- C3: Determine whether a test is one tailed or two tailed based on the claim.

General problem

A benchmark claim may refer either to a population proportion or to a population mean.

For a proportion setting, the claim states that specific percentage (for example, 62%) of the population has the characteristic of interest. A sample of number of samples (for example, 200) observations gives number of success observed successes (for example, 112). Use a significance level of significance percentage (for example, 5%).

For a mean setting, the claim states that the population mean is a benchmark mean value (for example, 75). A sample of number of samples (for example, 36) observations gives a sample mean value (for example, 79). If needed, the population standard deviation is given as a known value (for example, 12). Use a significance level of significance percentage (for example, 1%).

Questions/Tasks.

- (C1) Identify whether the setting is a proportion claim or a mean claim, and map the benchmark value, sample information, and significance level to the correct symbols.
- (C2) Write the null and alternative hypotheses for the three directions of the claim.
- (a) Write the null hypothesis as “parameter equals the benchmark” and the alternative as “parameter is lower than the benchmark”.

- (b) Write the null hypothesis as “parameter equals the benchmark” and the alternative as “parameter is higher than the benchmark”.
 - (c) Write the null hypothesis as “parameter equals the benchmark” and the alternative as “parameter is different from the benchmark”.
- (C3) Classify the tail type for each direction of the claim and justify it in words.

C1

- Identify whether the problem is about a population proportion or a population mean.
- For a proportion problem:
 - state that p is the true population proportion,
 - state that p_0 is the benchmark value (for example, 62%),
 - define n as the sample size (for example, 200),
 - define x as the observed count (for example, 112),
 - define α as the significance level (for example, 5%).
- For a mean problem:
 - state that μ is the true population mean,
 - state that μ_0 is the benchmark value (for example, 75),
 - define n as the sample size (for example, 36),
 - define $\bar{x}$ as the sample mean (for example, 79),
 - define σ as the known population standard deviation when it is provided (for example, 12),
 - define α as the significance level (for example, 1%).
- Match each symbol correctly to the numerical values given in the statement.
- Parts “lower”, “higher”, and “different” use the same parameter and benchmark, but different alternative claims.

C2

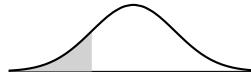
- Write the hypotheses correctly for each part (a), (b), and (c), and include α in each case.
- For a proportion problem, use p and p_0 :
 - (a) $H_0: p = p_0$, $H_a: p < p_0$, with the stated significance level. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is lower than the benchmark.
 - (b) $H_0: p = p_0$, $H_a: p > p_0$, with the stated significance level. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is higher than the benchmark.
 - (c) $H_0: p = p_0$, $H_a: p \neq p_0$, with the stated significance level. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is different from the benchmark.
- For a mean problem, use μ and μ_0 :
 - (a) $H_0: \mu = \mu_0$, $H_a: \mu < \mu_0$, with the stated significance level. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is lower than the benchmark.
 - (b) $H_0: \mu = \mu_0$, $H_a: \mu > \mu_0$, with the stated significance level. In words, the null hypothesis says that the

population mean equals the benchmark, while the alternative says that the population mean is higher than the benchmark.

- (c) $H_0: \mu = \mu_0$, $H_a: \mu \neq \mu_0$, with the stated significance level. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is different from the benchmark.
- In every case, keep the null hypothesis as “parameter equals the benchmark” and match the inequality direction to lower, higher, or different.

C3

- Classify the tail type for each direction of the claim and justify it in words.
 - (a) left-tailed test, because the claim says “lower”.



- (b) right-tailed test, because the claim says “higher”.



- (c) two-tailed test, because the claim says “different”.



- This mapping is the same for both proportion and mean tests.

1 Daily app check-ins for students and recent graduates

A university career center uses a mobile app to support students and recent graduates with internship alerts, interview reminders, and short productivity check-ins. The center is reviewing whether users are staying engaged enough for the app to justify next semester’s funding, especially because many users are balancing classes, part-time work, and job applications. The current benchmark says that 62% of active users complete the daily check-in. A class project reviews 200 active users and finds that 112 completed the daily check-in. They will use a 5% significance level for this check.

Questions/Tasks.

- (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value, the sample size, and the observed sample result for the daily check-in proportion to the symbols used in the solution.
- (C2) Write hypotheses for each case using the stated significance level.
- (a) Using a significance level of 5%, write the null and alternative hypotheses for the case where the true daily check-in proportion is lower than 62%.
 - (b) Using a significance level of 5%, write the null and alternative hypotheses for the case where the true daily check-in proportion is higher than 62%.
 - (c) Using a significance level of 5%, write the null and alternative hypotheses for the case where the true daily check-in proportion is different from 62%.
- (C3) For each case, classify the test as left-tailed, right-tailed, or two-tailed and justify using the wording of the alternative claim.

C1

- This is a population proportion setting.
- Let p be the true population proportion of active users who check in daily.
- Let p_0 be the benchmark proportion, with $p_0 = 62\%$.
- Let n be the sample size, with $n = 200$.
- Let x be the observed number of daily check-ins, with $x = 112$.
- Let α be the significance level, with $\alpha = 5\%$.
- Parts “lower”, “higher”, and “different” use the same parameter p and the same benchmark p_0 , but different alternative claims.

C2

- (a) $H_0: p = 0.62$, $H_a: p < 0.62$, $\alpha = 0.05$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is lower than the benchmark.
- (b) $H_0: p = 0.62$, $H_a: p > 0.62$, $\alpha = 0.05$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is higher than the benchmark.
- (c) $H_0: p = 0.62$, $H_a: p \neq 0.62$, $\alpha = 0.05$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is different from the benchmark.

C3

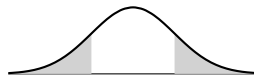
- (a) Left-tailed test because the alternative claims the parameter is lower than the benchmark.



- (b) Right-tailed test because the alternative claims the parameter is higher than the benchmark.



- (c) Two-tailed test because the alternative claims the parameter is different from the benchmark.



2 Evening study time during a demanding term

A university academic support office is evaluating whether students are spending enough time studying each evening during a demanding midterm period. Many of them are combining coursework with internships, long commutes, or part-time jobs, so the office wants to know whether the reported average still reflects current study habits. The report claims that students study an average of 75 minutes per evening. A sample of 36 students has an average of 79 minutes per evening. Assume the population standard deviation is known to be 12 minutes. They will use a 10% significance level for this check.

Questions/Tasks.

- (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value, the sample size, and the observed sample result for the mean study time to the symbols used in the solution.
- (C2) Write hypotheses for each case using the stated significance level.
- (a) Using a significance level of 10%, write the null and alternative hypotheses for the case where the true mean study time is lower than 75 minutes.
 - (b) Using a significance level of 10%, write the null and alternative hypotheses for the case where the true mean study time is higher than 75 minutes.
 - (c) Using a significance level of 10%, write the null and alternative hypotheses for the case where the true mean study time is different from 75 minutes.
- (C3) For each case, classify the test as left-tailed, right-tailed, or two-tailed and justify using the wording of the alternative claim.

C1

- This is a population mean setting.
- Let μ be the true population mean study time per evening for students.
- Let μ_0 be the benchmark mean, with $\mu_0 = 75$.
- Let n be the sample size, with $n = 36$.
- Let $\bar{x}$ be the sample mean study time, with $\bar{x} = 79$.
- Let σ be the known population standard deviation, with $\sigma = 12$.
- Let α be the significance level, with $\alpha = 10\%$.
- Parts “lower”, “higher”, and “different” use the same parameter μ and the same benchmark μ_0 , but different alternative claims.

C2

- (a) $H_0: \mu = 75$, $H_a: \mu < 75$, $\alpha = 0.10$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is lower than the benchmark.
- (b) $H_0: \mu = 75$, $H_a: \mu > 75$, $\alpha = 0.10$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is higher than the benchmark.
- (c) $H_0: \mu = 75$, $H_a: \mu \neq 75$, $\alpha = 0.10$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is different from the benchmark.

C3

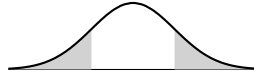
- (a) Left-tailed test because the alternative claims the parameter is lower than the benchmark.



- (b) Right-tailed test because the alternative claims the parameter is higher than the benchmark.



(c) Two-tailed test because the alternative claims the parameter is different from the benchmark.



3 Defect rate in portable chargers

A consumer electronics brand markets portable chargers to people who depend on their phones for maps, payment apps, job communication, and ride-hailing. After several customer complaints from users who said the chargers failed during commutes or long campus and work days, the quality team reviews whether the production line is still meeting its usual standard. The factory process has a benchmark defect proportion of 6%. In a quality sample of 400 chargers, 34 are defective. Because defect monitoring requires a stricter standard, they will use a 2% significance level for this check.

Questions/Tasks.

- (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value, the sample size, and the observed sample result for the defect proportion to the symbols used in the solution.
- (C2) Write hypotheses for each case using the stated significance level.
- Using a significance level of 2%, write the null and alternative hypotheses for the case where the true defect proportion is lower than 6%.
 - Using a significance level of 2%, write the null and alternative hypotheses for the case where the true defect proportion is higher than 6%.
 - Using a significance level of 2%, write the null and alternative hypotheses for the case where the true defect proportion is different from 6%.
- (C3) For each case, classify the test as left-tailed, right-tailed, or two-tailed and justify using the wording of the alternative claim.

C1

- This is a population proportion setting.
- Let p be the true population proportion of defective chargers from the process.
- Let p_0 be the benchmark defect proportion, with $p_0 = 6\%$.
- Let n be the sample size, with $n = 400$.
- Let x be the observed number of defective chargers, with $x = 34$.
- Let α be the significance level, with $\alpha = 2\%$.
- Parts “lower”, “higher”, and “different” use the same parameter p and the same benchmark p_0 , but different alternative claims.

C2

- (a) $H_0: p = 0.06$, $H_a: p < 0.06$, $\alpha = 0.02$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is lower than the benchmark.
- (b) $H_0: p = 0.06$, $H_a: p > 0.06$, $\alpha = 0.02$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is higher than the benchmark.
- (c) $H_0: p = 0.06$, $H_a: p \neq 0.06$, $\alpha = 0.02$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is different from the benchmark.

C3

- (a) Left-tailed test because the alternative claims the parameter is lower than the benchmark.



- (b) Right-tailed test because the alternative claims the parameter is higher than the benchmark.



- (c) Two-tailed test because the alternative claims the parameter is different from the benchmark.



4 Homework completion time in an evening certification program

A digital learning platform offers evening certification courses for people trying to strengthen their resumes while working or looking for their first full-time job. Because many users study late at night and have limited free time, the platform team is checking whether homework tasks are taking longer or shorter than expected and whether assignment design should be changed. The platform states that the average homework completion time is 48 minutes. A sample of 49 assignments gives an average of 45 minutes. Assume the population standard deviation is known to be 10 minutes. To be stricter because assignment design changes affect many users, they will use a 1% significance level for this check.

Questions/Tasks.

- (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value, the sample size, and the observed sample result for the mean completion time to the symbols used in the solution.
- (C2) Write hypotheses for each case using the stated significance level.
- (a) Using a significance level of 1%, write the null and alternative hypotheses for the case where the true mean completion time is lower than 48 minutes.
 - (b) Using a significance level of 1%, write the null and alternative hypotheses for the case where the true mean completion time is higher than 48 minutes.

- (c) Using a significance level of 1%, write the null and alternative hypotheses for the case where the true mean completion time is different from 48 minutes.
- (C3) For each case, classify the test as left-tailed, right-tailed, or two-tailed and justify using the wording of the alternative claim.

C1

- This is a population mean setting.
- Let μ be the true population mean homework completion time for all assignments on the platform.
- Let μ_0 be the benchmark mean, with $\mu_0 = 48$.
- Let n be the sample size, with $n = 49$.
- Let $\bar{x}$ be the sample mean completion time, with $\bar{x} = 45$.
- Let σ be the known population standard deviation, with $\sigma = 10$.
- Let α be the significance level, with $\alpha = 1\%$.
- Parts “lower”, “higher”, and “different” use the same parameter μ and the same benchmark μ_0 , but different alternative claims.

C2

- (a) $H_0: \mu = 48$, $H_a: \mu < 48$, $\alpha = 0.01$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is lower than the benchmark.
- (b) $H_0: \mu = 48$, $H_a: \mu > 48$, $\alpha = 0.01$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is higher than the benchmark.
- (c) $H_0: \mu = 48$, $H_a: \mu \neq 48$, $\alpha = 0.01$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is different from the benchmark.

C3

- (a) Left-tailed test because the alternative claims the parameter is lower than the benchmark.



- (b) Right-tailed test because the alternative claims the parameter is higher than the benchmark.



- (c) Two-tailed test because the alternative claims the parameter is different from the benchmark.

